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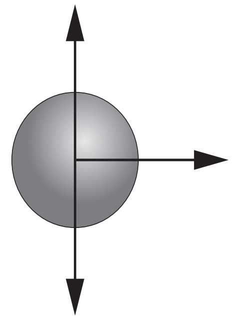
2. *(a)* The forces acting on a hailstone falling in a horizontal cross-wind can be represented as in the diagram.



(i) Calculate the magnitude and direction of the resultant force acting on the hailstone. [3]

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(ii) At a later time, the wind has stopped blowing and the hailstone falls at terminal velocity. In terms of forces, explain why the hailstone is at terminal velocity. [1]

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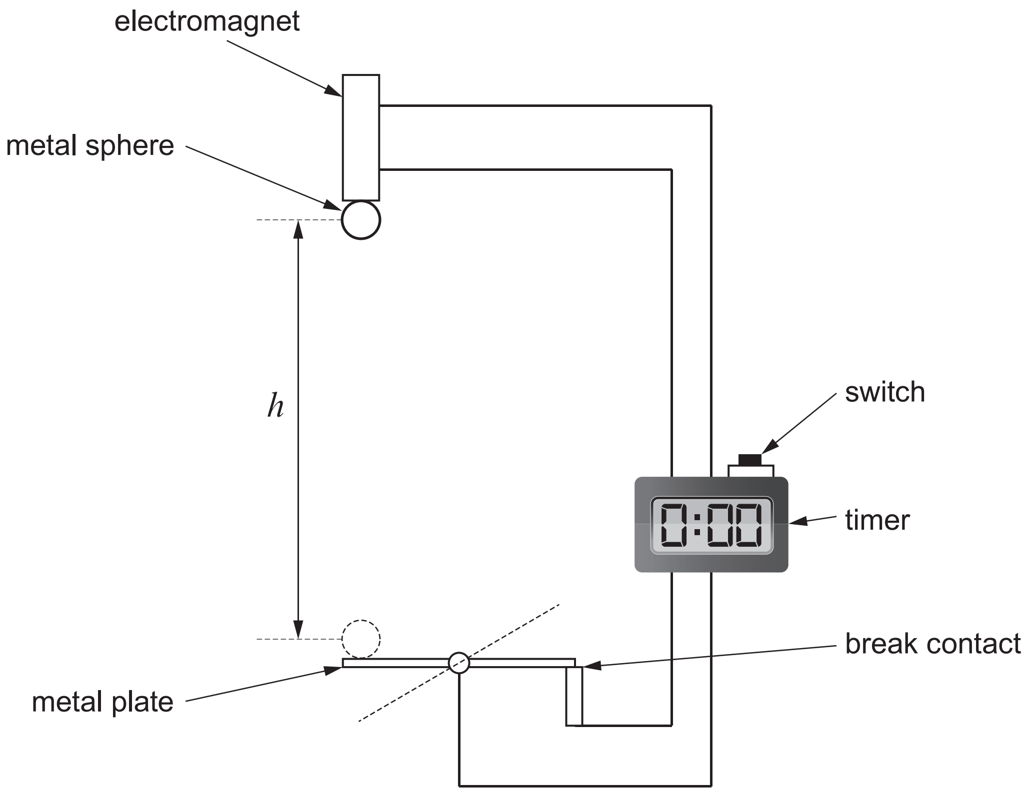
(b) Aled uses the following apparatus to measure the acceleration of free-fall, g.



When the switch is pressed, it starts the timer and disconnects the electromagnet, almost instantly releasing the metal sphere. When the sphere hits the metal plate it breaks the circuit, stopping the timer. The time taken for the metal sphere to fall through a range of different heights, h is measured.

(i) Aled is told that there is a very small time delay between the switch being pressed and the ball being released. This is a systematic error. The manufacturer states that the time delay is 0.05 s. State how Aled should account for the systematic error when taking readings. [1]

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(ii) Aled records his **corrected results (i.e. with the systematic error accounted for)** in the table below. Complete the row for time squared, t^2 giving your answers to an appropriate number of significant figures. [2]

Drop height, h/m	0.40	0.80	1.20	1.60	2.00
Corrected time, t/s	0.27	0.41	0.48	0.58	0.64
Corrected time squared, t^2/s^2					

(iii) The following relationship is used to find a value for g:

$$g = \frac{2h}{t^2}$$

Show how this relationship is obtained from an appropriate equation of accelerated motion. [2]

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(iv) On the grid below, plot a graph of h **(vertical axis)** against t^2 **(horizontal axis)** and draw a line of best fit. [4]

(v) Use your graph to determine a value for g. [3]

(c) Discuss to what extent your graph agrees with the equation in *(b)*(iii). [3]

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